

Furey Fermions in GALG and FFGFT: N-Operator, Tripotents, and Neutrino-Vacuum Identity

Abstract

Doug Matzke's IPI mail of 3 September 2026 presents concrete GALG results on Furey fermions in $Cl(6)$: complete Up/Down fermion sets Su/Sd , idempotents N_1, N_2, N_3 , tripotent N , jE7 projector, and a grade-1+grade-5 construction. The strongest single result: $Su[0] = Vss[0]$ exactly — the neutrino is the lowest vacuum state **[B]**. Theorem F shows: the tripotent property for even vacua and the Frobenius fixed-point property (Doc. 339) are the same decomposition from two independent directions **[B]**. The electron exception at the jE7 projector is the known Furey problem **[S]**.

.1 Theorem A — N_k are idempotents in characteristic 3

$N_k = -1 + (-i B_k)$, $B_k^2 = -1$ (Witt pair):

$$N_k^2 = 1 + 2i B_k + (-1)(-1) = 2 + 2i B_k \equiv -1 + (-i B_k) \pmod{3} = N_k. \quad \mathbf{[B]}$$

The sum loses its scalar part since $-3 \equiv 0$: $N_1 + N_2 + N_3 \equiv (-i) \sum B_k = N$.

.2 Theorem B — N is a tripotent

Cross-terms $N_k N_j$ ($k \neq j$) are pure grade-4 elements; hence $N^3 = N$ **[B]**. Doug's output: $N**3 = N$, $N**4 = N**2$.

.3 Theorem C — $Su[0] = Vss[0]$: neutrino is the vacuum

$Su[0] = Vss[0]$: True and $(1+N)*(1+jE7) = Vss[0]$: True **[B]**.

FFGFT correspondence [B]. In FFGFT, v_1 lies in orbit $O_1 = \{1, 3, 9\}$, the orbit of the generator of $GF(27)^*$ — the algebraically simplest object. Both frameworks independently place the neutrino at the algebraically simplest position: convergence of two independent frameworks.

.4 Theorems D–F

D — sigma [B]. $e_1 \rightarrow e_2 \rightarrow e_4 \rightarrow e_1$, $e_3 \rightarrow e_6 \rightarrow e_5 \rightarrow e_3$; $\{1:2, 2:4, 4:1, 3:6, 6:5, 5:3\}$, two 3-cycles, $\sigma^3 = \text{id}$. Doc. 341 contains this formula; Doug's `all_g1_g5good` is the σ -invariant sum.

E — $jE7 \equiv N_k$ as projectors [B]; electron exception [S]. $F \cdot jE7 = +F$ for all Su (8/8); $F \cdot jE7 = -F$ for 7/8 of Sd (Sd[7]=electron fails). Same for N_k . Known Furey problem. FFGFT: electron in Dirac layer $\{f_3, f_4\}$, neutrino in Majorana layer $\{f_1, f_2\}$ (Doc. 343, Theorem D'''). Common root open [S].

F — Tripotent for even vacua [B]. $(V_k + P)^2 = -V_k$, $(V_k + P)^3 = V_k$ for $k \in \{0, 2, 4\}$; no collapse for $k \in \{1, 3, 5\}$ (length 32). Identical to the Frobenius fixed-point/orbit split of Doc. 339 — two independent derivations of the same decomposition.

.5 Summary

Table 1: Results Doc. 344

Theorem	Content	Status
A	$N_k^2 = N_k$, characteristic-3 proof	[B]
B	$N^3 = N$, tripotent	[B]
C	Su[0]=Vss[0]: neutrino=vacuum; FFGFT correspondence	[B]
D	$\text{sigma} = \{1 \rightarrow 2, 2 \rightarrow 4, 4 \rightarrow 1, 3 \rightarrow 6, 6 \rightarrow 5, 5 \rightarrow 3\}$	[B]
E	$jE7 \equiv N_k$; electron exception (Furey problem)	[B]/[S]
F	Tripotent even vacua = Frobenius fixed points (Doc. 339)	[B]

Open question [S]: Common algebraic root of the electron inconsistency in GALG/Furey and the Majorana/Dirac layer separation in FFGFT (Doc. 343, Theorem D''').

Verification script

pruef_344_furey_galg.py — 6 theorems, all assertions passed.

Bibliography

- [1] C. Furey, *Standard Model Physics from an Algebra?* (2016).
- [2] D. Matzke, IPI mail 3 September 2026.
- [3] J. Pascher, *Doc. 339: Frobenius separation*, FFGFT corpus (Aug. 2026).
- [4] J. Pascher, *Doc. 341: GF(27) in GALG and FFGFT*, FFGFT corpus (Sept. 2026).
- [5] J. Pascher, *Doc. 343: Zeta function of T^4/Z_3* , FFGFT corpus (Sept. 2026).